Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3301-WE01

Title: **MEng/BEng:** Structures and Geomatics 3		
Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Question 1

(a) By analysing the beam shown in Figure Q1.1 using the geometric stiffness approach, determine the 1st and 2nd critical elastic buckling loads. The flexural stiffness EI is constant over the entire length of the beam.

Hint: Ignore the axial deformation of the beam.

[40%]

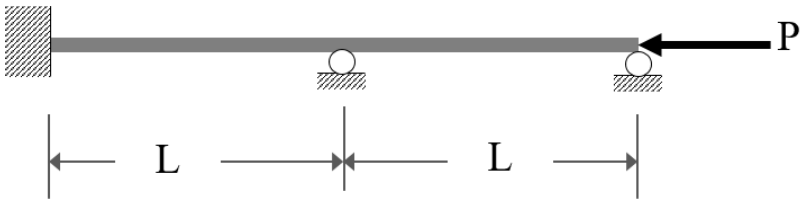


Figure Q1.1

(b) Figure Q1.2 shows a continuous beam structure comprising three spans (3m, 4m and 5m) with a uniformly distributed load in span CD of 21.6 kN/m. Steel sections are used throughout with the same cross section used in BC and CD and a different cross section in AB. The moment of inertia in span AB is equal to half the value in span BC and span CD (i.e. $I_{BC} = I_{CD} = 2\ I_{AB}$). Carry out a moment distribution analysis of the beam taking the following steps:

(i) Determine the distribution factors

[10%]

(ii) Determine the fixed end moments

[10%]

(iii) Determine the balance moments at the support (work out the whole numbers and carry out a maximum of five balancing iterations)

[30%]

(iv) Find the value of the vertical reaction at C

[10%]

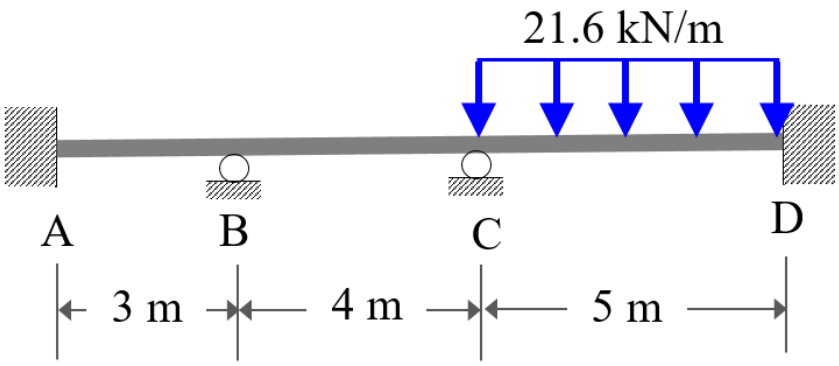


Figure Q1.2

Continued

Question 2

Consider the continuous beam in Figure Q2.1 with a fixed support at the left end, a roller support at the right, and an intermediate roller support and uniformly distributed loading as shown. Assume that the material and section properties are constant throughout the length of the beam and neglect axial loads and deformation.

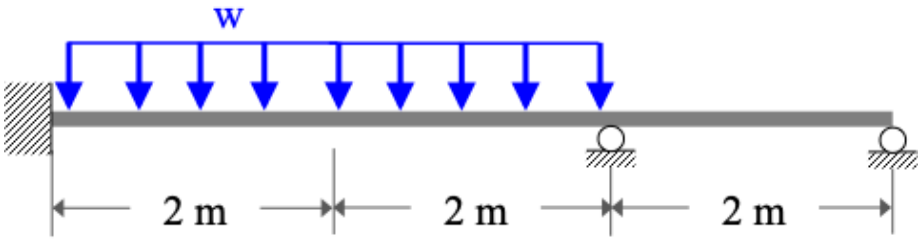


Figure Q2.1

(a) Sketch the expected mode of failure, and calculate the value of the uniformly distributed load *w* that would lead to collapse of the beam due to the formation of plastic hinge(s) as a function of the plastic moment (M_p).

[35%]

(b) Based on the assumed mode of failure for *Part (a)* estimate the support reactions and sketch the quantitative (i.e., labeling all minima and maxima) shear force and bending moment diagrams for the beam at the point of collapse as a function of the plastic moment (M_p).

[35%]

Now, consider the frame shown in Figure Q2.2 with pin supports at the bottom, rigid connections between each beam and column, and uniformly distributed vertical loading and horizontal concentrated loading as shown. Assume that the material and section properties are constant throughout.

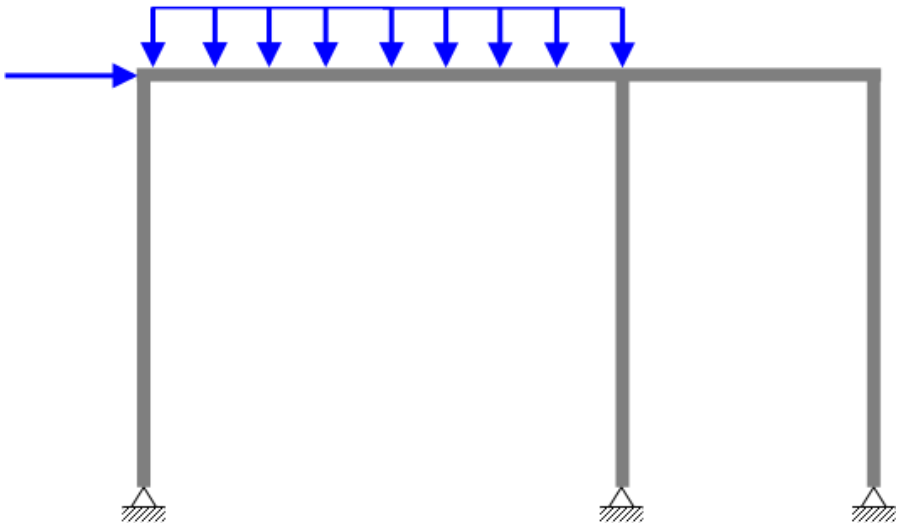


Figure Q2.2

(c) Describe, including sketches, three possible collapse mechanisms and explain the procedure to determine the critical loading based on these collapse mechanisms.

[30%]

Continued

Question 3

a) Electronic Distance Measurement (EDM)

A slope distance from station A to station B has been measured using EDM. Two frequencies have been used, the data from the measurement is shown in table Q3.1.

Table Q3.1

Frequency	Wavelength	Phase difference
1	10.0m	0.867π
2	11.11m	1.381π

Determine the slope distance in metres. Then determine the horizontal and vertical distance if the vertical angle from station A to station B is 16°45'36".

[40%]

b) Earthworks Volumes

A cutting is to be excavated for a straight road running through a hill composed of clay soil. The values of eight consecutive cross-sectional areas taken at a constant horizontal interval of 20m are shown in Table Q3.2.

Table Q3.2. Section area data.

	CS1	CS2	CS3	CS4	CS5	CS6	CS7	CS8
Chainage	00	20	40	60	80	100	120	140
Cross sectional area (m^2)	156	134	145	123	164	148	120	101

Once excavated the clay will be transported off site in standard ($10m^3$ capacity) wagons. Determine the volume of material that will need to be transported and hence the number of wagon journeys required.

[30%]

(c) Errors in Levelling

A line levelling survey has been carried out between two Ordnance Survey Benchmarks as show in Table Q3.3. Using the data provided complete the arithmetic checks for the Rise and Fall method and calculate the misclosure. Prove this misclosure is unacceptable and detail some potential sources for this error and how the errors could be mitigated in future surveys.

Continued

Table Q3.3. Levelling Survey Booking Form

Staff Position	Back Sight	Fore Sight	Rise	Fall	Reduced Level	Remarks / Arithmetic Checks
BM1	0.347				45.125	BM on Student Union (45.125m)
CP1	0.536	2.367		2.020	43.105	
CP2	3.108	1.086		0.550	42.555	
BM2		1.816	1.292		43.847	BM on Engineering Building (43.834m)
SUM						
Checks						
Misclosure						

[30%]

END